$\sum_{k=1}$ $\sum_{k=1}$ $\sum_{k=1}$

Example 30: *If $a_1, a_2, a_3, \ldots, a_n$ are real numbers, then show that*

$$\left(\sum_{k=1}^{n} a_k\right)^2 \le n\sum_{k=1}^{n} a_k^2$$

Solution: If we put $b_k = 1 (1 \le k \le n)$ in the Cauchy-Schwarz inequality $\left(\sum_{k=1}^{n} a_k b_k\right)^2 \le \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right)$, we have

$$\left(\sum_{k=1}^{n} a_k\right)^2 \le \left(\sum_{k=1}^{n} a_k^2\right)(1+1+1+\ldots+1) = n\sum_{k=1}^{n} a_k^2 \Rightarrow \left(\sum_{k=1}^{n} a_k\right)^2 \le n\sum_{k=1}^{n} a_k^2$$

Example 31: *If a_1, a_2, a_3, a_4, a_5 are real numbers, then show that*

$$(a_1 + a_2 + a_3 + a_4 + a_5)^2 \le 5[a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2]$$

Solution: Cauchy-Schwarz inequality for five real numbers is

$$\left(\sum_{k=1}^{5} a_k b_k\right)^2 \le \left(\sum_{k=1}^{5} a_k^2\right)\left(\sum_{k=1}^{5} b_k^2\right) \quad \ldots(1)$$

If we put $b_k = 1 (1 \le k \le 5)$, i.e. $b_1 = 1 = b_2 = b_3 = b_4 = b_5$ in (1), then

$$\left(\sum_{k=1}^{5} a_k\right)^2 \le \left(\sum_{k=1}^{5} a_k^2\right)(1+1+1+1+1) = 5\sum_{k=1}^{5} a_k^2 \Rightarrow \left(\sum_{k=1}^{5} a_k\right)^2 \le 5\sum_{k=1}^{5} a_k^2$$

$$\Rightarrow (a_1 + a_2 + a_3 + a_4 + a_5)^2 \le 5[a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2]$$

Example 32: *If $\underline{x} = (x_1, x_2, \ldots, x_n), \underline{y} = (y_1, y_2, \ldots, y_n) \in R^n$, then show that $d(\underline{x},\underline{y}) = \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}\right|$ is a metric on R^n.*

Solution: M_1): $d(\underline{x},\underline{y}) = \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}\right| \ge 0, \forall \underline{x},\underline{y} \in R^n$

M_2): $d(\underline{x},\underline{y}) = 0 \Leftrightarrow \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}\right| = 0 \Leftrightarrow (x_i - y_i)^2 = 0$

$$\Leftrightarrow x_i = y_i (1 \le i \le n) \Leftrightarrow \underline{x} = \underline{y}$$

$$M_3): d(\underline{x},\underline{y}) = \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}\right| = \left|\sqrt{\sum_{i=1}^{n}(y_i - x_i)^2}\right| = d(\underline{y},\underline{x})$$

$$M_4): d(\underline{x},\underline{z}) = \left|\sqrt{\sum_{i=1}^{n}(x_i - z_i)^2}\right| = \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i + y_i - z_i)^2}\right|$$

Let $a_i = x_i - y_i, b_i = y_i - z_i$, then the last equaion takes the form

$$d(\underline{x},\underline{z}) = \left|\sqrt{\sum_{i=1}^{n}(a_i + b_i)^2}\right| \qquad \ldots(1)$$

Using Minkosk's inequality, we have

$$\sqrt{\sum_{i=1}^{n}(a_i + b_i)^2} \le \sqrt{\sum_{i=1}^{n}a_i^2} + \sqrt{\sum_{i=1}^{n}b_i^2}$$

$$\Rightarrow \left|\sqrt{\sum_{i=1}^{n}(a_i + b_i)^2}\right| \le \left|\sqrt{\sum_{i=1}^{n}a_i^2} + \sqrt{\sum_{i=1}^{n}b_i^2}\right| \le \left|\sqrt{\sum_{i=1}^{n}a_i^2}\right| + \left|\sqrt{\sum_{i=1}^{n}b_i^2}\right|$$

Using this in (1), we have

$$d(\underline{x},\underline{z}) = \left|\sqrt{\sum_{i=1}^{n}(a_i + b_i)^2}\right| \le \left|\sqrt{\sum_{i=1}^{n}a_i^2}\right| + \left|\sqrt{\sum_{i=1}^{n}b_i^2}\right|$$

$$\Rightarrow d(\underline{x},\underline{z}) \le \left|\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}\right| + \left|\sqrt{\sum_{i=1}^{n}(y_i - z_i)^2}\right|$$

$$\Rightarrow d(\underline{x},\underline{z}) \le d(\underline{x},\underline{y}) + d(\underline{y},\underline{z}),\ \forall\ \underline{x},\underline{y},\underline{z} \in R^n$$

Example 33: *Show that the space l_2 of all real sequences $x = \{x_n\}$ such that $\sum_{k=1}^{\infty}|x_k|^2 < \infty$ is a metric space.*

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Solution: The distance function d on l_2 is defined as follows:

$$d(x,y) = \sqrt{\sum_{k=1}^{\infty}|x_k - y_k|^2}$$

$M_1)$: $d(x,y) = \sqrt{\sum_{k=1}^{\infty}|x_k - y_k|^2} \ge 0,\ \forall x,y \in l_2$

$M_2)$: $d(x,y) = 0 \Leftrightarrow \sqrt{\sum_{k=1}^{\infty}|x_k - y_k|^2} = 0 \Leftrightarrow \sum_{k=1}^{\infty}|x_k - y_k|^2 = 0$

$$\Leftrightarrow |x_k - y_k|^2 = 0\ (k \in N)$$

$$\Leftrightarrow |x_k - y_k| = 0 \ (k \in N)$$

$$\Leftrightarrow x_k - y_k = 0 \ (k \in N)$$

$$\Leftrightarrow x_k = y_k = 0 \ (k \in N)$$

$$\Leftrightarrow x = y$$

$M_3)$: $d(x,y) = \sqrt{\sum_{k=1}^{\infty} |x_k - y_k|^2} = \sqrt{\sum_{k=1}^{\infty} |y_k - x_k|^2} = d(y,x)$

$M_4)$: $d(x,z) = \sqrt{\sum_{k=1}^{\infty} |x_k - z_k|^2} = \sqrt{\sum_{k=1}^{\infty} |x_k - y_k + y_k - z_k|^2}$

Let $a_k = x_k - y_k$, $b_k = y_k - z_k$, then the last equation takes the form

$$d(x,z) = \sqrt{\sum_{k=1}^{\infty} |a_k + b_k|^2} \qquad \ldots(1)$$

Using Minkosk's inequality, we have

$$\sqrt{\sum_{k=1}^{\infty} |a_k + b_k|^2} \le \sqrt{\sum_{k=1}^{\infty} |a_k|^2} + \sqrt{\sum_{k=1}^{\infty} |b_k|^2}$$

Using this in (1), we have

$$d(x,z) = \sqrt{\sum_{k=1}^{\infty} |a_k + b_k|^2} \le \sqrt{\sum_{k=1}^{\infty} |a_k|^2} + \sqrt{\sum_{k=1}^{\infty} |b_k|^2}$$

$$\Rightarrow d(x,z) \le \sqrt{\sum_{k=1}^{\infty} |x_k - y_k|^2} + \sqrt{\sum_{k=1}^{\infty} |y_k - z_k|^2}$$

$$\Rightarrow d(x,z) \le d(x,y) + d(y,z), \qquad \forall x, y, z \in l_2$$

This shows that l_2 is a metric space